



September 2019 Examination

EC400

Probability and Statistical Inference

Instructions to candidates

Time allowed: 2 hours

This paper contains **SEVEN** questions. Answer **ALL** questions.

On your script, write

- Your name
- Your class group number
- The name of your class teacher.

You are not allowed to use an electronic calculator

1. **(8 points - 10 minutes)** *I brought various bags with tulip bulbs from the Netherlands for my friends. Each bag contains 25 tulip bulbs. Three-fourths of the bags contain bulbs for 5 red and 20 yellow tulips; one fourth of the bags contain bulbs for 15 red and 10 yellow tulips. A bag is selected at random and one bulb is planted. Give the probability that it will produce a*

(a) *Red tulip, say $P(R)$.*

Introduce a new random variable B such that $B = 1$ if one selects a bag that contains 5 red and 20 yellow tulips, and $B = 2$ if one selects a bag that contains 15 red and 10 yellow.

$$\begin{aligned} P(R) &= P(\text{draw a red bulb} | B = 1) * P(B = 1) + P(\text{draw a red bulb} | B = 2) * P(B = 2) \\ &= \frac{5}{25} * \frac{3}{4} + \frac{15}{25} * \frac{1}{4} = \frac{3}{20} + \frac{3}{20} = \frac{3}{10} \end{aligned}$$

4 points.

- (b) *If the tulip is red, find the conditional probability that a bag having 15 red and 10 yellow tulips was selected.*

Using Bayes' Theorem:

$$P(B = 2 | R) = \frac{P(R | B = 2)P(B = 2)}{P(R)} = \frac{\frac{15}{25} * \frac{1}{4}}{\frac{3}{10}} = 1/2$$

4 points. Give 2 when Bayes' rule is stated correctly. (give full marks if incorrect answer from a) is used but the rest is good.)

2. **(8 points - 10 minutes)** *Let X have the p.d.f. $f(x) = x^2/9$, $0 < x < 3$, zero elsewhere. Find the p.d.f. of $Y = X^3$. What is the median of the distribution of Y ?*

2 points: First of all we note that over its domain it is a one-to-one transformation.

With $u(x) = x^3$, the inverse is given by $v(y) = \sqrt[3]{y}$

5 points: $f_y(y) = f_x(v(y)) \cdot |v'(y)|$ for $0 < y < 27$. With $|v'(y)| = \frac{1}{3}y^{-2/3}$, we obtain

$$\begin{aligned} f_y(y) &= \frac{(y^{1/3})^2}{9} \times \frac{1}{3}y^{-2/3} = \frac{1}{27} \text{ for } 0 < y < 27 \\ &= 0 \text{ elsewhere} \end{aligned}$$

(students loose 3 points for omitting relevant ranges)

1 point: The median of the distribution of Y is 13.5.

3. **(13 points - 15 minutes)** The joint probability density function of X and Y is given by

$$f(x, y) = 2e^{-x-y}, 0 \leq y \leq x < \infty$$

- (a) Find the marginal p.d.f. of X .

$$\int_0^x 2e^{-x-y} dy = 2e^{-x} \int_0^x e^{-y} dy = 2e^{-x} (1 - e^{-x}) \text{ for } 0 \leq x < \infty$$

3 points: clearly states how the marginal pdf is obtained

2 points: derivation of the marginal pdf.

Subtract 2 points for incorrect use of integral boundaries.

- (b) Find the conditional p.d.f. of Y given $X = x$.

$$f(y|X = x) = \frac{f(y, x)}{f(x)} = \frac{2e^{-x-y}}{2e^{-x}(1 - e^{-x})} = \frac{e^{-y}}{1 - e^{-x}} \text{ for } 0 \leq y \leq x < \infty$$

3 points: clearly states the rule of obtaining conditional pdf

2 points: executing the rule properly (given the answers obtained - even if they were wrong)

1 points: clearly indicating ranges

- (c) Are X and Y independent?

The fact that in b) the conditional distribution depends on x , tells us that X and Y are not independent.... A proof showing $f(x, y) \neq f_x(x)f_y(y)$ takes more time but should receive full credit.

2 points for any correct answer.

4. **(12 points - 14 minutes)** Let X_1 and X_2 be continuous random variables. If X_1 and X_2 are independent it can be shown that $E(g(X_1)h(X_2)) = E(g(X_1))E(h(X_2))$ where g and h are continuous functions.

- (a) Discuss how we can obtain $E(g(X_1)h(X_2))$ in the setting where X_1 and X_2 are not independent using the joint density of X_1 and X_2 .

[3 points] This is standard bookwork

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x_1)h(x_2) \underbrace{f_{X_1, X_2}(x_1, x_2)}_{\text{joint density}} dx_2 dx_1$$

- (b) *An alternative method to obtain this expectation uses the law of iterated expectations. In view of this, show how you can obtain this expectation using the conditional distribution $f_{X_2|X_1=x_1}$ and the marginal distribution f_{X_1} instead. Clearly explain the intuition and validity of the law of iterated expectations.*

[2 points] The law of iterated expectations states

$$E(g(X_1)h(X_2)) = E_{X_1}[E(g(X_1)h(X_2)|X_1 = x_1)]$$

[2 points] The law of iterated expectations makes use of the fact that

$$f_{X_1, X_2}(x_1, x_2) = f_{X_2|X_1=x_1}(x_2|X_1 = x_1)f_{X_1}(x_1)$$

[2 points] Some further technical details revealing validity. Plugging this in the above relation gives:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x_1)h(x_2) \underbrace{f_{X_2|X_1=x_1}(x_2|X_1 = x_1)f_{X_1}(x_1)}_{\text{product conditional * marginal}} dx_2 dx_1$$

Rewriting allows us to write this as

$$\begin{aligned} & \int_{-\infty}^{\infty} g(x_1)f_{X_1}(x_1) \left[\underbrace{\int_{-\infty}^{\infty} h(x_2)f_{X_2|X_1=x_1}(x_2|X_1 = x_1) dx_2}_{E(h(X_2)|X_1=x_1)} \right] dx_1 \\ &= \int_{-\infty}^{\infty} [g(x_1)E(h(X_2)|X_1 = x_1)] f_{X_1}(x_1) dx_1 \\ &= E_{X_1}[g(x_1)E(h(X_2)|X_1 = x_1)] \\ &= E_{X_1}[E(g(X_1)h(X_2)|X_1 = x_1)] \end{aligned}$$

[3 points] Intuition: This tells us to that evaluate $E(g(X_1)h(X_2))$, we first treat X_1 as fixed (by conditioning on $X_1 = x_1$), and deal with the stochastic nature of X_1 after first evaluating $E(g(X_1)h(X_2)|X_1 = x_1) \equiv g(x_1)E(h(X_2)|X_1 = x_1)$

5. **(25 points - 30 minutes)** *The random variable X is normally distributed with mean μ and variance $\sigma^2 = 100$. We want to test the null hypothesis $H_0 : \mu = 10$ against the alternative $H_1 : \mu \neq 10$ at the 5% level of significance. A sample of $n = 100$ independent realizations of X was collected.*

- (a) *Discuss how you can obtain the probability of committing a type-II error if the true value of μ is 10.5.*

[7 points] The $\Pr(\text{type II error}) = \Pr(\text{Not reject } H_0 | \mu = 10.5) = \Pr\left(-1.96 \leq \frac{\bar{X}-10}{1} \leq 1.96 | \mu = 10.5\right) = \Pr\left(-2.46 \leq \frac{\bar{X}-10.5}{1} \leq 1.46\right)$ where $\frac{\bar{X}-10.5}{1} \sim N(0, 1)$ – Observe that $\sigma^2/n = 1$. Therefore $\Pr(\text{type II error}) = \Phi(1.46) - \Phi(-2.46) = .9278 - .0069 = 0.9209$. This signals a low power of our test, $\text{Power} = 1 - \Pr(\text{type II error}) = 0.0791$.

- i. *What happens to this probability if the variance σ^2 becomes smaller and/or the sample n becomes bigger? Discuss.*

[3 points] If σ^2/n goes down (either σ^2 smaller or n larger) the precision of our estimator increases. This will increase our test statistic $\frac{\bar{X}-10}{\sigma/\sqrt{n}}$ and will allow us to reject earlier, which improves the power of our test and consequently lower the $\Pr(\text{type II error})$

- ii. *What happens to this probability if you had considered the one-sided alternative $H_1' : \mu > 10$ at the 5% level of significance instead? Discuss.*

[3 points] Student may want to draw, signalling to the fact that we start rejecting earlier in the area where it matters. Therefore we also start to reject earlier when the null is false, e.g., $\mu = 10.5$, and the power increases, or $\Pr(\text{type II error})$ goes down.

- (b) *What is the p-value of the test if $\bar{x} = 10.22$?*

[6 points] The p-value gives us the lowest significance at which we want to reject the null. It is given by $\Pr\left(\left|\frac{\bar{X}-10}{1}\right| > \frac{10.22-10}{1}\right) = 2\Phi(-0.22) = 2 * .4129 = 0.8258$

- i. *What happens to this p-value if the variance σ^2 becomes smaller and/or the sample n becomes bigger? Discuss.*

[3 points] Since the p-value is $\Pr\left(\left|\frac{\bar{X}-10}{\sigma/\sqrt{n}}\right| > \frac{10.22-10}{\sigma/\sqrt{n}}\right) = 2\Phi\left(-\frac{0.22}{\sigma/\sqrt{n}}\right)$ we see that a decrease in σ^2/n will make $-\frac{0.22}{\sigma/\sqrt{n}}$ more negative and the p-value goes down.

- ii. *What happens to this p-value if you had considered the one-sided alternative $H_1' : \mu > 10$, instead? Discuss.*

[3 points] The p-value then is half of the two sided value as we reject only on one side.

6. **(12 points - 15 minutes)** *The random variable X is said to have a $\chi^2(r)$ distrib-*

ution. Its moment generating function is given by

$$M(t) = (1 - 2t)^{-r/2} \text{ with } t < 1/2.$$

- (a) Use the moment generating function to show that the mean and variance of the random variable X equal to r and $2r$ respectively.

3 points: proper statement about relationship between MGF and moments

2 points: proper execution of formulae

The MGF satisfies the property that $E(X^k) = M^{(k)}(0)$. To obtain the mean, we need to evaluate $M'(0)$

$$M'(t) = \frac{\partial M(t)}{\partial t} = -\frac{r}{2} (1 - 2t)^{-r/2-1} \times -2 = r (1 - 2t)^{-r/2-1}$$

$$M'(0) = r \Rightarrow \text{The mean of } X : E(X) = r$$

To obtain the variance, we recall: $\text{Var}(X) = E(X^2) - (E(X))^2 = E(X^2) - r^2$.

$$M''(t) = \frac{\partial^2 M(t)}{\partial^2 t} = r \left(-\frac{r}{2} - 1 \right) (1 - 2t)^{-r/2-2} \times -2 = (r^2 + 2r) (1 - 2t)^{-r/2-1}$$

$$M''(0) = r^2 + 2r$$

$$\Rightarrow \text{the variance of } X : \text{Var}(X) = r^2 + 2r - (r)^2 = 2r$$

- (b) Based on a random sample $X_1, \dots, X_n$, provide two method of moment estimators for r . Clearly describe the intuition behind this estimator.

Hint: Note that you have two moment conditions involving r .

4 points: proper understanding of MME as revealed by answer and discussion of MME estimators

3 points: proper execution of formula (subtract one point if student doesn't realize that estimate for variance cannot be negative).

The first MME uses $E(X) = r$: The MME chooses $\hat{r}$ to ensure that its sample analogue is satisfied:

$$\frac{1}{n} \sum_{i=1}^n X_i = \hat{r}_{MME}$$

(the sample average is the sample analogue of the population mean).

The second MME uses $E(X^2) = r^2 + 2r$:

$$\begin{aligned}\frac{1}{n} \sum_{i=1}^n X_i^2 &= \hat{r}_{MME}^2 + 2\hat{r}_{MME} \\ \hat{r}_{MME}^2 + 2\hat{r}_{MME} - \frac{1}{n} \sum_{i=1}^n X_i^2 &= 0 \\ \Rightarrow \hat{r}_{MME} &= \frac{-2 \pm \sqrt{4 + 4 \frac{1}{n} \sum_{i=1}^n X_i^2}}{2} = -1 \pm \sqrt{1 + \frac{1}{n} \sum_{i=1}^n X_i^2}\end{aligned}$$

To ensure answer is positive (variance): $\hat{r}_{MME} = \frac{-2 \pm \sqrt{4 + 4 \frac{1}{n} \sum_{i=1}^n X_i^2}}{2} = -1 + \sqrt{1 + \frac{1}{n} \sum_{i=1}^n X_i^2}$

7. **(22 points - 26 minutes)** Let $X_1, \dots, X_n$ be a random sample from distribution with density function given by

$$f(x) = \frac{x}{\theta^2} e^{-x/\theta}, \quad 0 < x < \infty, \text{ and } 0 < \theta < \infty.$$

For this random variable $E(X) = 2\theta$ and $\text{Var}(X) = 2\theta^2$.

- (a) Show that the MLE estimator for θ equals $\bar{X}/2$ where $\bar{X}$ is the sample average.
Advice: Do not overspend your time here, ensure you clarify and annotate the steps required. You do not need the derivations to answer the rest of this question.

By independence the likelihood can be expressed as

$$\begin{aligned}L(\theta) &= \prod_{i=1}^n f(x_i) \\ &= (1/\theta^2)x_1 e^{-x_1/\theta} \dots (1/\theta^2)x_n e^{-x_n/\theta} = \frac{1}{\theta^{2n}} x_1 \dots x_n e^{-\frac{1}{\theta} \sum_{i=1}^n x_i} \\ \ln L(\theta) &= -2n \ln(\theta) + \sum_{i=1}^n \ln(x_i) - \frac{1}{\theta} \sum_{i=1}^n x_i \\ \text{F.O.C.: } \frac{\partial \ln L}{\partial \theta} &: -\frac{2n}{\hat{\theta}_{MLE}} + \frac{1}{\hat{\theta}_{MLE}^2} \sum_{i=1}^n x_i = 0 \Rightarrow \hat{\theta}_{MLE} = \frac{1}{2n} \sum_{i=1}^n x_i \\ &\Rightarrow \hat{\theta}_{MLE} = \frac{1}{2} \bar{X}\end{aligned}$$

3 points: provide the log-likelihood function (subtract 1 point if use of independence is not clearly referred to)

3 points: FOC (properly using $\hat{\theta}_{MLE}$) and solve (subtract 1 point if θ is simply used)

- (b) *Discuss whether the MLE estimator for θ is unbiased and/or consistent. Explain what we understand by concepts of unbiasedness and consistency.*

4 point: The estimator is unbiased: $E\left(\hat{\theta}_{MLE}\right) = \frac{1}{2n} \sum_{i=1}^n E(X_i) = \frac{1}{2n} \sum_{i=1}^n 2\theta = \theta$.

2 points: Unbiasedness is a property that ensures that under repeated sampling, our estimator on average is correct.

3 points: The estimator is consistent: $\text{plim } \hat{\theta}_{MLE} = \text{plim } \frac{1}{2n} \sum_{i=1}^n X_i = \frac{1}{2} \text{plim } \frac{1}{n} \sum_{i=1}^n X_i = \frac{1}{2} E(X_i) = \theta$. We have used **Slutsky** and **Khinchine LLN**. [Obviously students could also have used the sufficient conditions, recognizing that the unbiasedness together with the fact that $\text{Var}(\hat{\theta}_{MLE}) =_{\text{indep}} \frac{1}{4} \frac{2\theta^2}{n} = \frac{\theta^2}{2n} \rightarrow 0$ as $n \rightarrow \infty$]

2 points: Consistency is a property that ensures that as our sample size is large, our estimator converges in probability to the truth - "gets closer to the truth".

- (c) *What would be a good approximation of the sampling distribution of the MLE estimator for θ ? What is the relevance of this result?*

Hint: You may assume that regularity conditions are satisfied so that we can rely on the CLT **2 points:** The CLT (Lindeberg-Levy) ensures that $\bar{X} \overset{a}{\sim} N(E(\bar{X}), \text{Var}(\bar{X}))$ (the sampling distribution of sums/averages can be well approximated by that of a normal distribution).

1 point: Here $\bar{X} \overset{a}{\sim} N(2\theta, 2\theta^2/n)$.

1 point: Since our estimator $\hat{\theta}_{MLE} = \frac{1}{2}\bar{X} \Rightarrow$

$$\hat{\theta}_{MLE} \overset{a}{\sim} N\left(\theta, \frac{\theta^2}{2n}\right)$$

2 points: The important of this result is that we can use it for hypothesis testing purposes.